AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION

MAY 2011

Mathematics Paper I

Time allowed: 40 minutes Marks 30

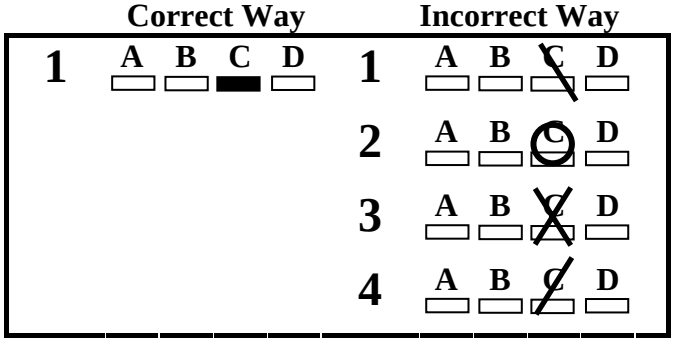
INSTRUCTIONS

1. Read each question carefully.

2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.

3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 30 only.

4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the box for your choice with a pencil as shown below.



5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new box.

6. DO NOT write anything in the answer grid. The computer only records what is in the boxes.

7. You may use a simple calculator if you wish.

1. If $3x-6=24$, then x is equal to

   A. 6
   B. 10
   C. 15
   D. 27

2. The Highest Common Factor (H.C.F) of $3a$, $6a^2$ and $9a^3$ is

   A. $3a$
   B. $6a$
   C. $3a^2$
   D. $6a^2$

3. To make $x^2+\frac{1}{4}$ a perfect square, we should add

   A. 1
   B. $\frac{x}{2}$
   C. x
   D. $2x$

4. Which one of the following is an improper algebraic fraction?

   A. $\frac{x^2+1}{x^3-1}$
   B. $\frac{(x+1)^2}{x^4+1}$
   C. $\frac{1}{x^2-1}$
   D. $\frac{(x+1)^2}{x-1}$

5. On simplification of $\frac{x^2-y^2}{x-y}\div(x+y)$ we get

   A. 0
   B. 1
   C. $(x+y)$
   D. $(x+y)^2$

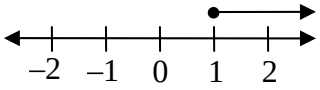
6. If $|x-d|=0$, then value of x is equal to

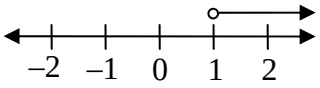
A. $-d$
B. d
C. d and $-d$
D. 0 and d

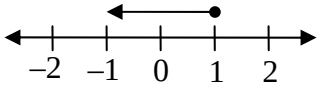
7. The solution of inequality $2-x>2$ is

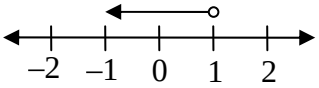
A. $x>-4$
B. $x<-4$
C. $x<0$
D. $x>0$

8. If x is a real number, then the solution of $\frac{3}{2}x-\frac{3}{2}>0$ can be represented on number line as

A. 

B. 

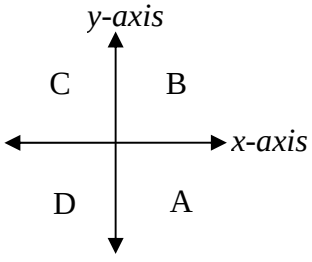
C. 

D. 

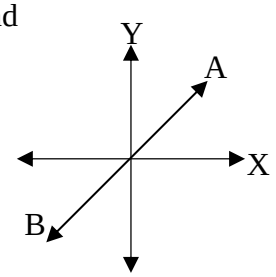
9. The point of intersection of straight lines $y=3x$ and $y=2x$ is

A. $(3,2)$
B. $(2,3)$
C. $(-3,-2)$
D. $(0,0)$

10. In the following figure, the 4th quadrant is represented by the letter



11. The pair of points which may lie on the straight line $\overleftrightarrow{AB}$ is $(0,0)$ and



A. $(2,0)$
B. $(3,-2)$
C. $(-3,-3)$
D. $(-2,2)$

12. The standard form of quadratic equation $x^2 - \frac{x}{3} = 1$ is

A. $x^2 - x - 3 = 0$
B. $3x^2 - x - 3 = 0$
C. $3x^2 - 3x + 3 = 0$
D. $3x^2 - x + 3 = 0$

13. The solution set of $x^2 = 5$ is

A. $\{\sqrt{5}\}$
B. $\{0, \sqrt{5}\}$
C. $\{\sqrt{5}, -\sqrt{5}\}$
D. $\{0, \pm\sqrt{5}\}$

14. The solution set of $\sqrt{x} = x$ is

A. $\{0\}$
B. $\{1\}$
C. $\{-1\}$
D. $\{0,1\}$

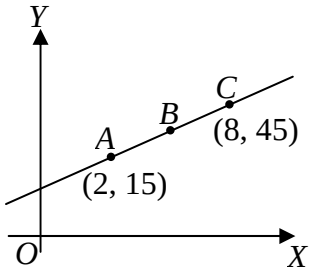
15. The distance between points (0, 0) and (1, 1) is

A. $-\sqrt{2}$ unit.
B. -1 unit.
C. 1 unit.
D. $\sqrt{2}$ unit.

16. If (2, 1) is the mid point of (2, 0) and (a, 2), then a is equal to

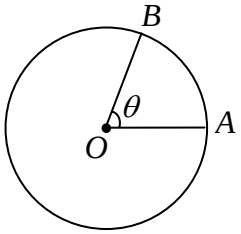
A. -2
B. 0
C. 1
D. 2

17. In the given diagram, point B is equidistant from points A and C. The coordinates of the point B are



A. (3, 15)
B. (5, 30)
C. (6, 30)
D. (10, 60)

18. In the given circle, lengths of arc AB and radius $\overline{OA}$ are equal. The measurement of angle θ is

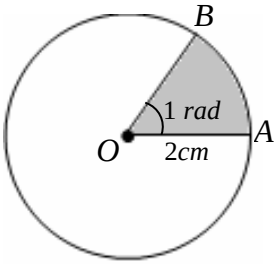


A. 1 radian.
B. 2 radian.
C. π radian.
D. 2π radian.

19. In radian measure, $60°$ is equal to

A. π radian.
B. $\frac{\pi}{2}$ radian.
C. $\frac{\pi}{3}$ radian.
D. $\frac{\pi}{4}$ radian.

20. The area of sector AOB in the given diagram is

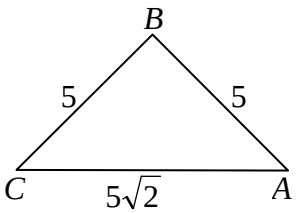


A. $1cm^2$.
B. $2cm^2$.
C. $4cm^2$.
D. $5cm^2$.

21. $\sqrt{2}\cos 45°$ is equal to

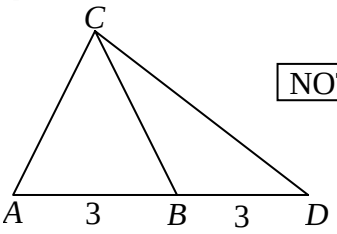
A. $\frac{1}{2}$
B. $\frac{1}{\sqrt{2}}$
C. 1
D. 2

22. In the following right angled triangle, length of hypotenuse and right angle are



A. 5 and $\angle B$.
B. $5\sqrt{2}$ and $\angle A$.
C. $5\sqrt{2}$ and $\angle B$.
D. 5 and $\angle A$.

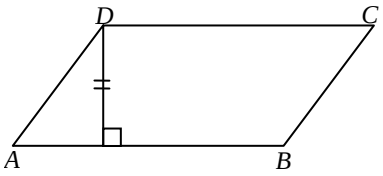
23. Consider the following figure. If the area of ΔABC is equal to $6cm^2$, then the area of ΔBDC is

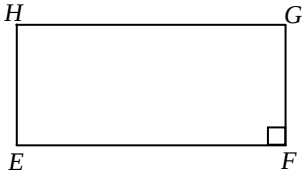


NOT TO SCALE

A. $6cm^2$.
B. $9cm^2$.
C. $12cm^2$.
D. $18cm^2$.

24. The areas of parallelograms *ABCD* and rectangle *EFGH* are equal. If both have the same altitude, then





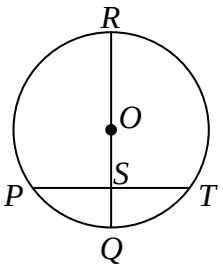
I. $AB \cong EF$
II. $AD \cong EH$
III. $CD \cong GH$

A. I only
B. II only
C. I and III only
D. II and III only

25. How many circles may pass through three non-collinear points?

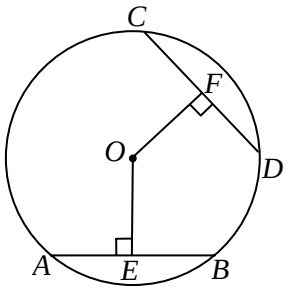
A. One
B. Two
C. Three
D. Infinite

26. In the given circle, if chord $\overline{PT}$ and diameter $\overline{QR}$ are intersecting each other, such that $m\overline{PS} = m\overline{TS}$, then $m\angle RST$ is



A. $60°$.
B. $80°$.
C. $90°$.
D. $120°$.

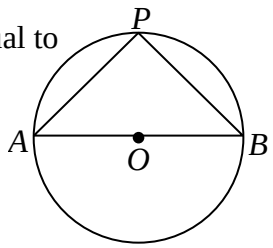
27. If *O* be the centre of the circle and $\overline{AB} \cong \overline{CD}$, then which one of the following statements is true?



I. $\overline{OE} \cong \overline{OF}$
II. $\overline{AE} \cong \overline{EB}$
III. $\overline{CF} \cong \overline{EB}$

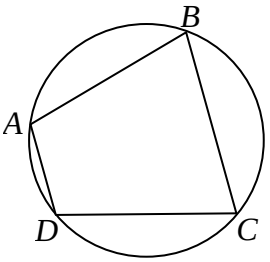
A. I, II and III
B. I only
C. II only
D. III only

28. In the given diagram, $\overline{AB}$ is the diameter of the circle, $m\angle APB$ is equal to



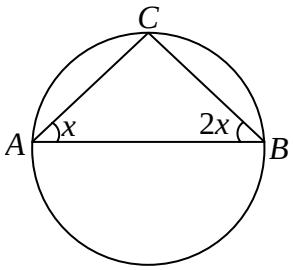
A. $45°$.
B. $60°$.
C. $90°$.
D. $120°$.

29. Which one of the following is true for the given figure?



A. $m\angle A + m\angle B = 90°$
B. $m\angle A + m\angle B = 180°$
C. $m\angle A + m\angle C = 90°$
D. $m\angle A + m\angle C = 180°$

30. If AB is the diameter of the given circle, then x is equal to



A. $30°$.
B. $45°$.
C. $60°$.
D. $90°$.

END OF PAPER

Please use this page for rough work